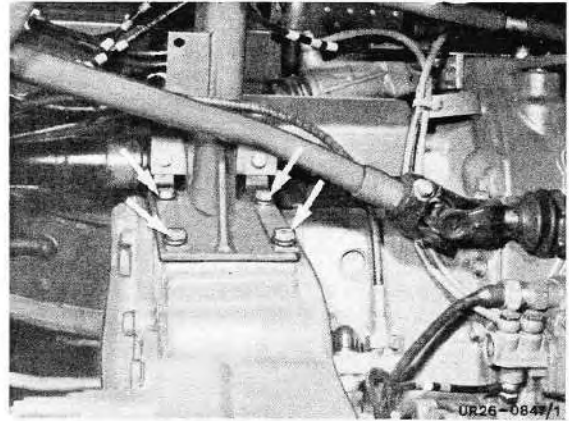
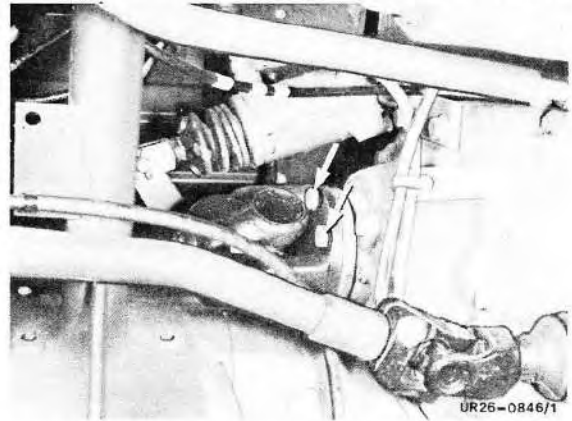


Sealing, Main Transmission Installed

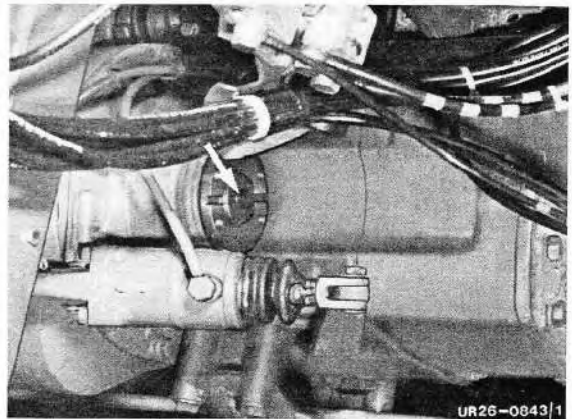
- 1 Raise driver's cab, see 60.5-2.1/1 or 60.5-2.2/1.
- 2 Detach sheetmetal cover.



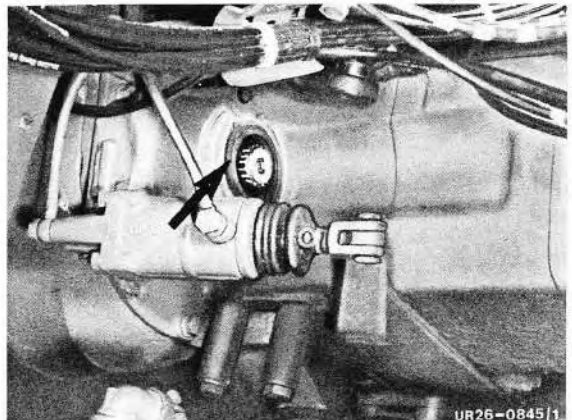
- 3 Detach input shaft.



- 4 Unscrew fastening bolt and remove drive flange.



- 5 Remove shaft sealing rings.

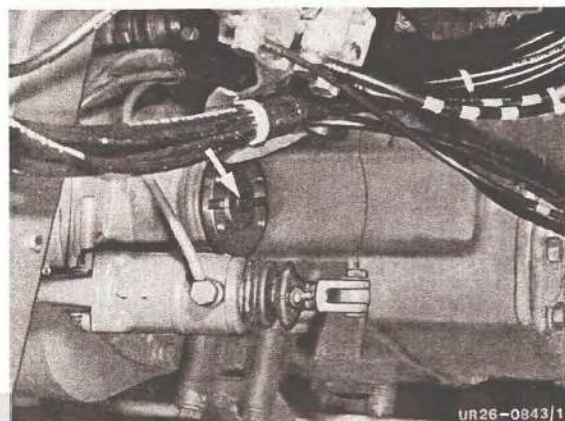


26.13 Sealing Oil Pump

717.9
SA 35 977

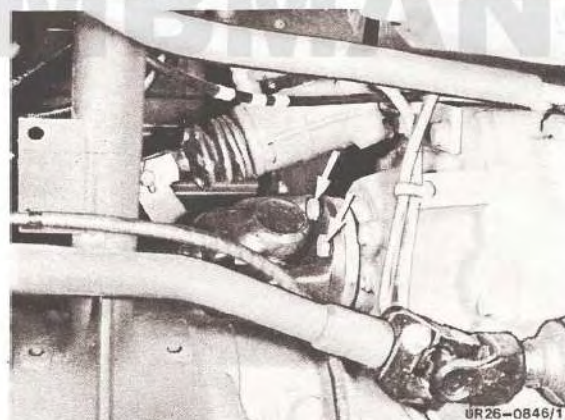


6 Press both shaft sealing rings into position using special tool No. 5.

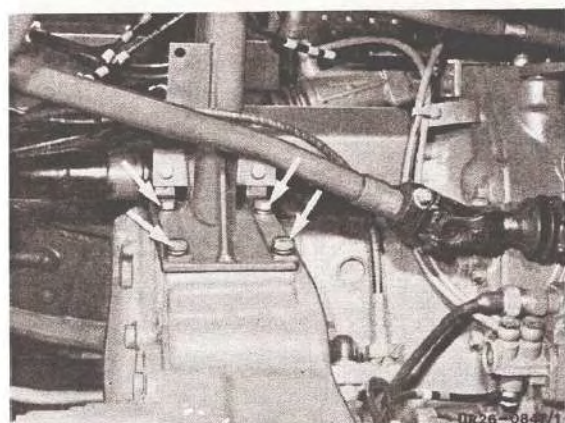


7 Press drive flange into position and tighten retainer together with lock washer and sealing ring.

See 1.7/1 for permissible tightening torques.



8 Attach input shaft.
Coat bolts with locking compound No. 1 and screw in.



9 Fit safety support and sheetmetal cover.

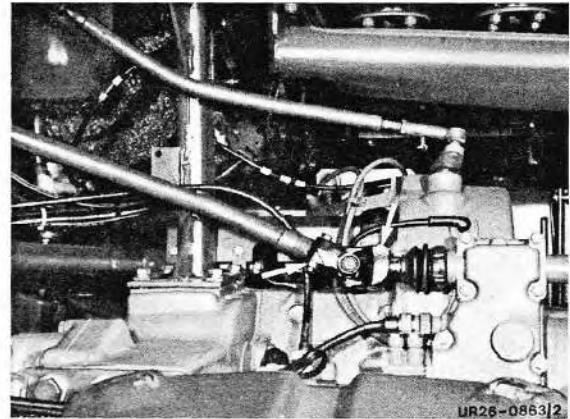
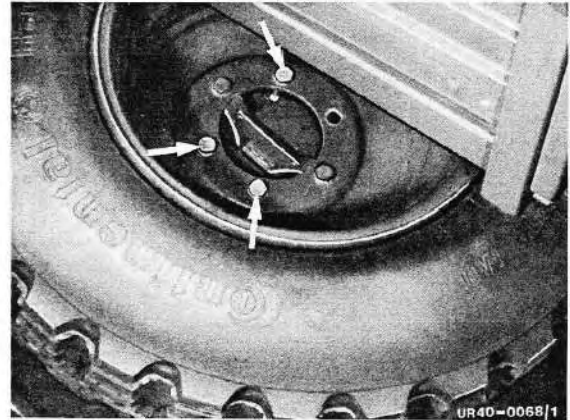
10 Lower driver's cab, see 60.5-2.1/1 or 60.5-2.2/1.

Adjusting Main Shift Lever

1 Remove spare wheel.

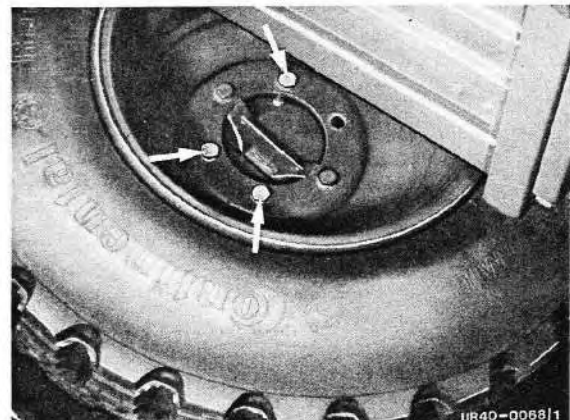
2 With main selector lever in 0 - position in the shift gate select 1st/2nd speed and release bolts at universal joint.

3 Adjust main shift lever and tighten bolts at universal joint.
For adjustments, see 1.5/1.



Adjusting Forward/Reverse Shift Lever

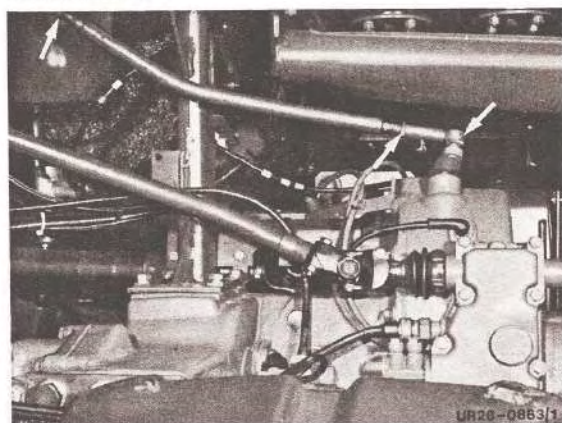
1 Remove spare wheel.



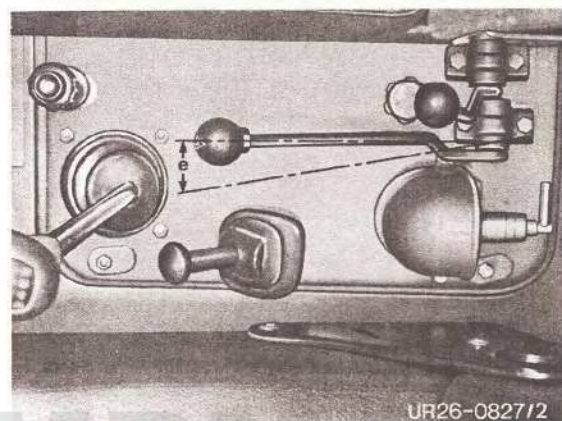
26.13 Adjusting Shift Lever

717.901

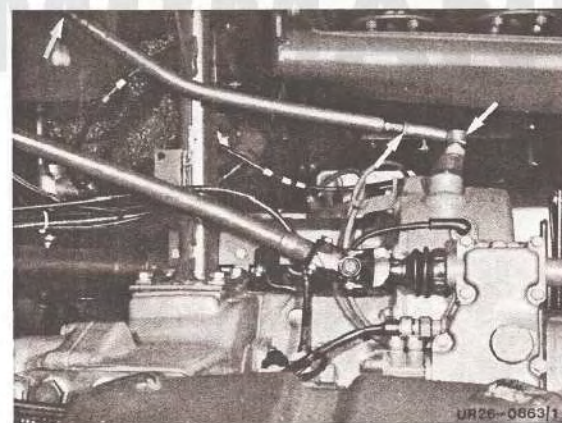
SA 35 977



2 Move shift lever to centre position and release nuts on gear shift linkage, detach shift linkage.



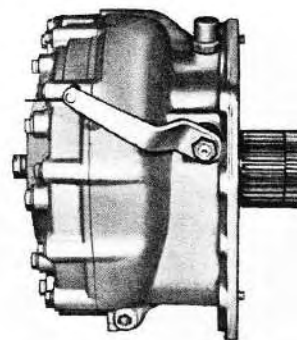
3 Adjust shift lever, while turning gear shift linkage to the left or right.
For adjustments, see 1.2/1.



4 Attach shift linkage and tighten nuts.

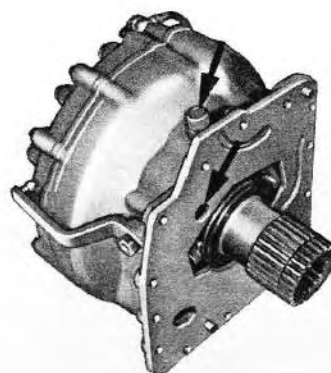
Disassembly, low ratio gears

- 1 Remove auxiliary transmission.
- 2 Mark the selector shaft for low ratio in the "engaged" position.



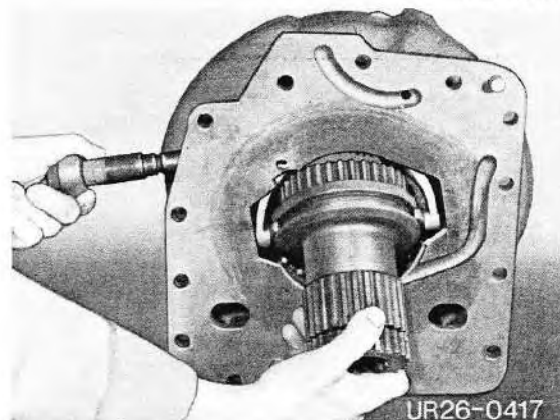
UR 26-1000/2

- 3 Remove the dowel pin in the low ratio selector fork and unscrew the interlock.



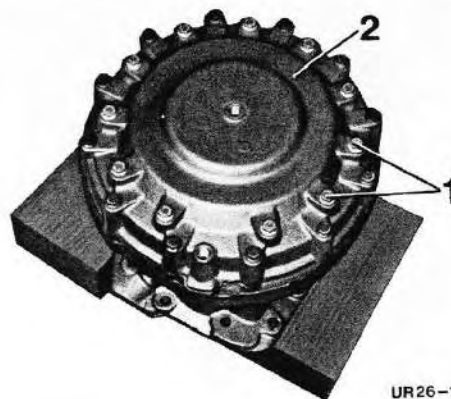
UR 26-1001/1

- 4 Remove selector shaft and sliding sleeve together with selector fork and rod.
- 5 Turn auxiliary transmission through 90°.
- 6 Remove oil pipe from cover plate.



UR26-0417

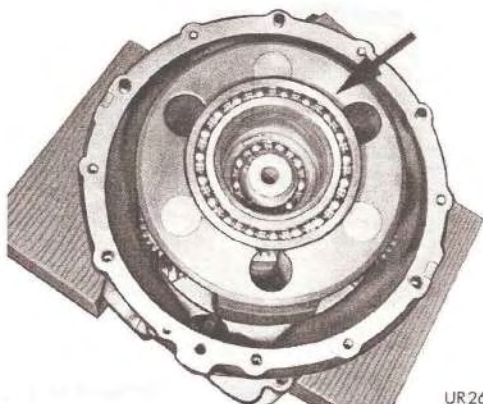
- 7 Loosen bolts for annulus (1) and remove cover plate (2).



UR26-1002/1

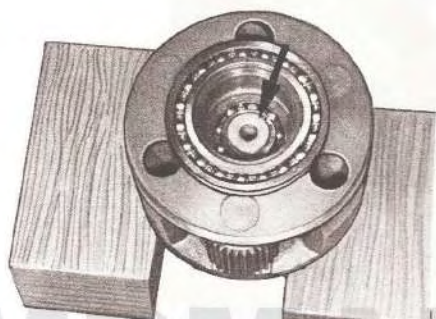
SA 35 736

- 8 Remove planetary gearing for low ratio gears.



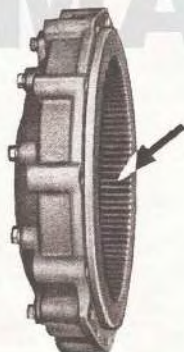
UR26-1003/1

- 9 Release propellor shaft and remove.
10 Release deep groove ball bearing and remove.



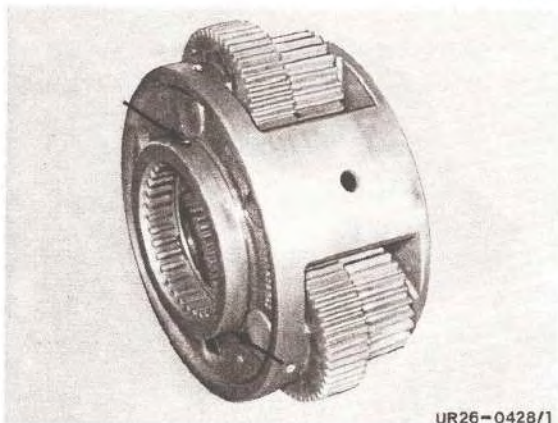
UR 26-1004/1

- 11 Remove annulus.



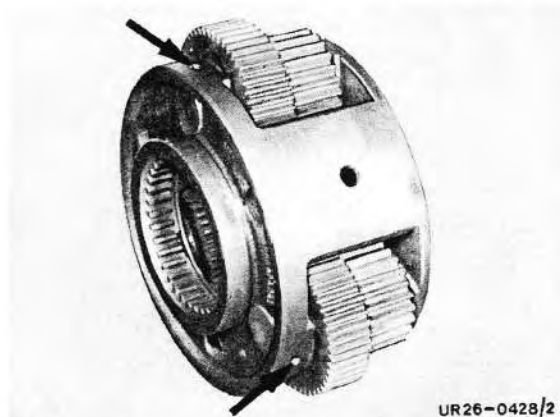
UR 26-1006/1

- 12 Unscrew bolts and remove ring gear.
Applicable from gearbox No. 003 168.



UR26-0428/1

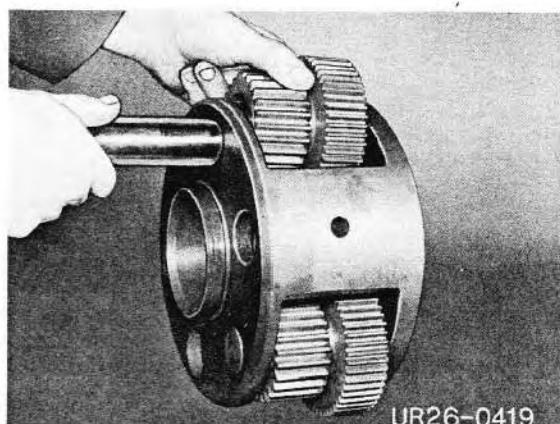
13 Release and unscrew bolt.



14 Remove planet shaft with planet pinion.

Note: Do not lose thrust washers, needle rollers and shims.

15 Clean and check all parts, renew as required. Blow out oil bores and oil pipes with compressed air.

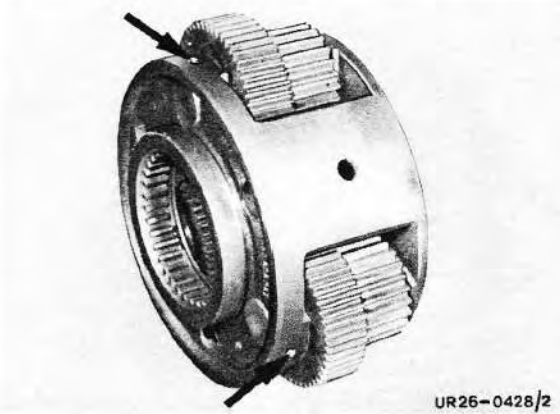


Assembly

1 Install planet pinion and planet shaft with thrust washers, needle rollers and shims.

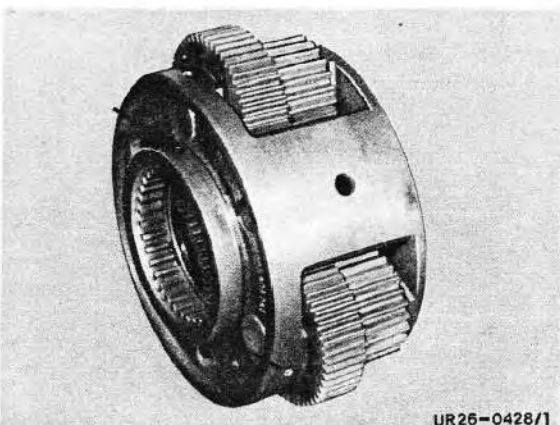
Note: Coat needle rollers with grease No. 5, renew needle rollers only in sets of 50.

2 Screw in bolts and secure with blow of chisel.



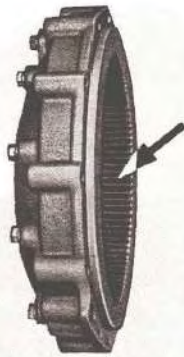
3 Install serrated washer, secure with bolt and press on large deep-groove ball bearing.

Note: Coat bolts with locking compound No. 1
Valid from transmission No. 003 168.



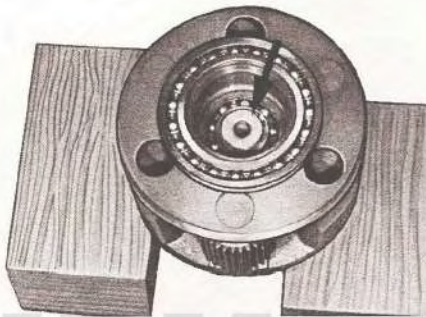
26.13 Disassembly and Assembly of Rear-Mounted Auxiliary Transmission

SA 35 736



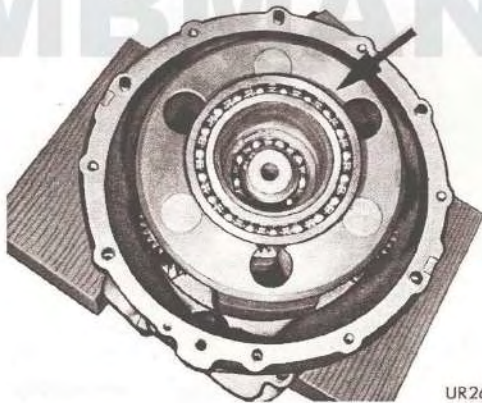
UR26-1006/1

- 4 Install ring gear.
- 5 Install deep-groove ball bearing and secure.



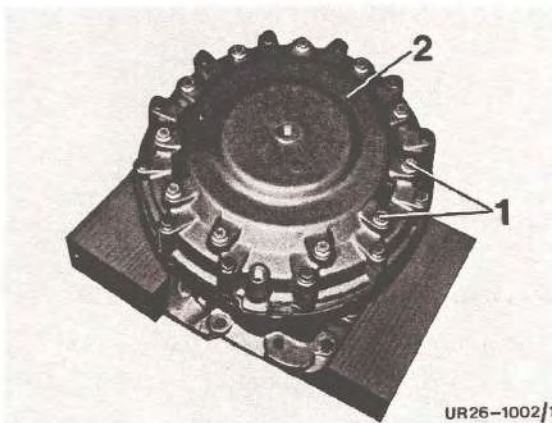
UR26-1004/1

- 6 Install input shaft and secure.



UR26-1003/1

- 7 Install planetary gear train, implement group.



UR26-1002/1

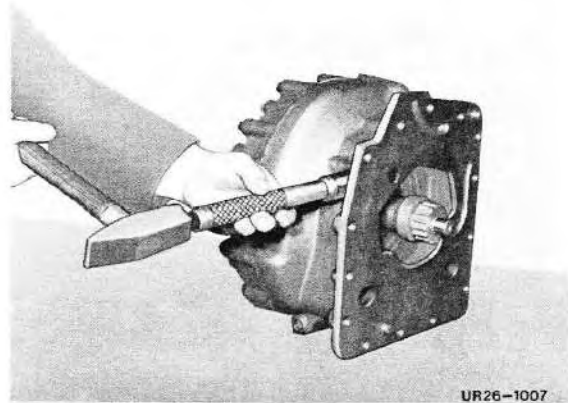
- 8 Fit end cover together with gasket, tighten bolts for ring gear (1) and end cover (2)
For permissible tightening torques, see 1.7/1.

- 9 Attach oil pipe to end cover.

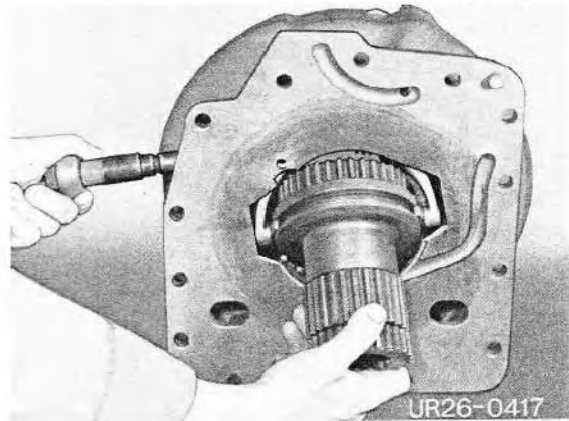
- 10 Turn rear-mounted auxiliary transmission through 90°.

Note: From chassis end No. 102 873 normal bolts are installed as standard in place of locking bolts.

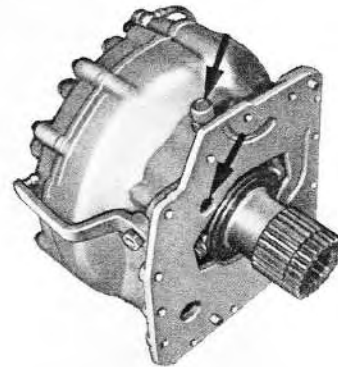
- 11 Coat shaft seal with grease No. 3 and fit with special tool No. 16a.



- 12 Fit selector fork with pads, sliding sleeve and selector shaft according to the markings.

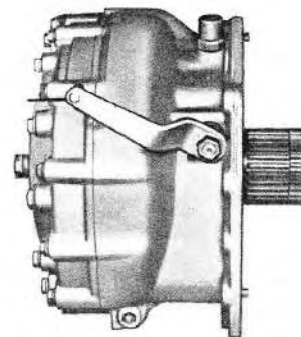


- 13 Fit dowel pin and interlock.



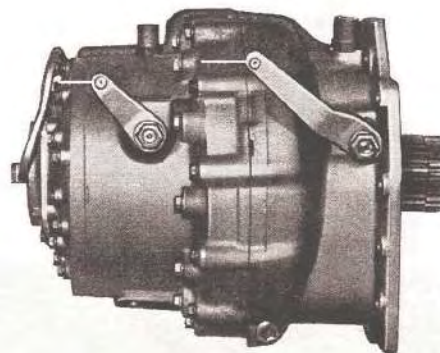
- 14 Select low ratio gears once and check selector lever position. If necessary, loosen nut and adjust selector lever, tighten and secure.
Measurement a, see 1.5/1.

- 15 Install auxiliary transmission with gasket.



Disassembly (low ratio and crawler gears)

- 1 Remove auxiliary transmission.
- 2 Mark the selector shaft for low ratio and crawler gears in the "engaged" position.



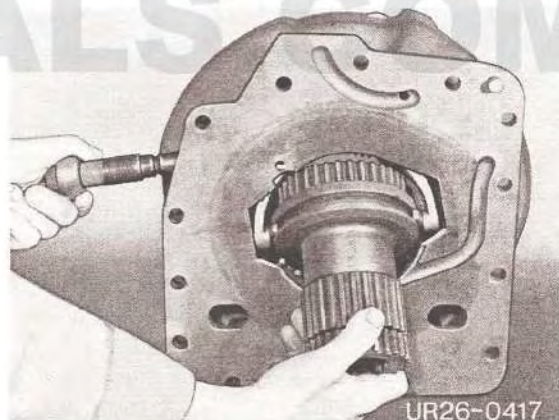
UR26-1008/2

- 3 Remove dowel pin in the low ratio selector fork and unscrew the interlock.



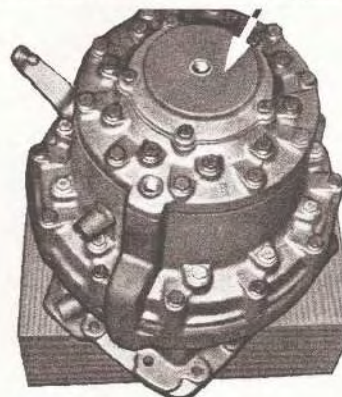
UR26-1009/1

- 4 Remove selector shaft and sliding sleeve together with selector fork and pads.
- 5 Turn auxiliary transmission through 90°.
- 6 Remove oil pipe on the crawler gear cover plate.

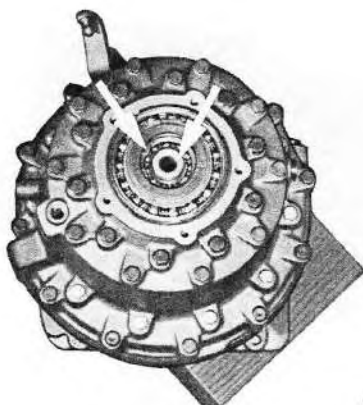


UR26-0417

- 7 Remove cover plate.

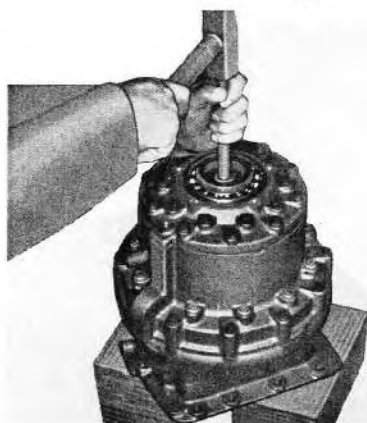


UR26-1010/1



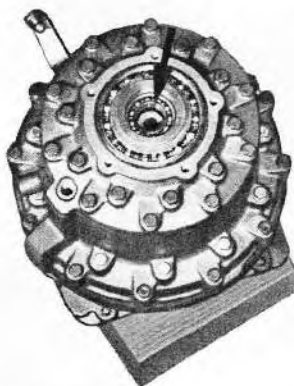
UR26-1011/1

- 8 Remove inner and outer circlips.



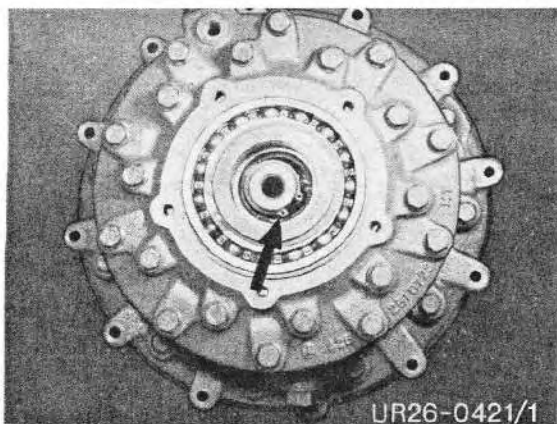
UR26-1012

- 9 Knock in input shaft.



UR26-1013/1

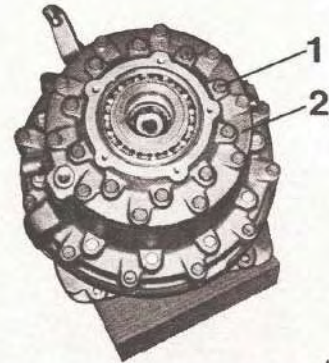
- 10 Pull out small deep groove ball bearing.



UR26-0421/1

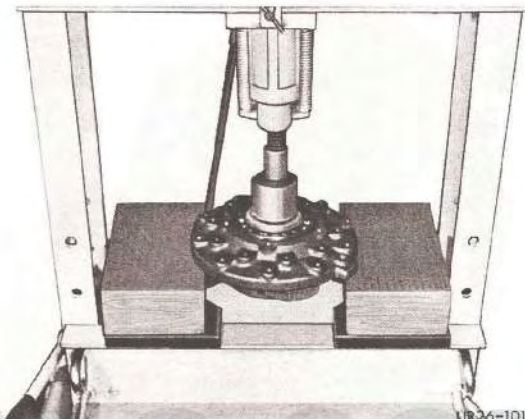
- 11 Remove circlips.

- 12 Loosen annulus (1) and remove bolts for crawler gear planetary gearing (2).



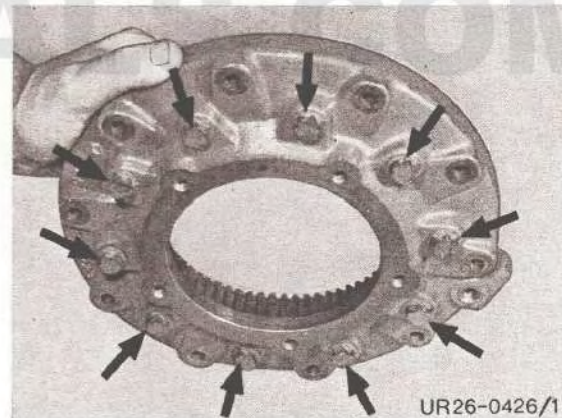
UR26-1023/1

- 13 Release crawler gear planetary gearing and press out.



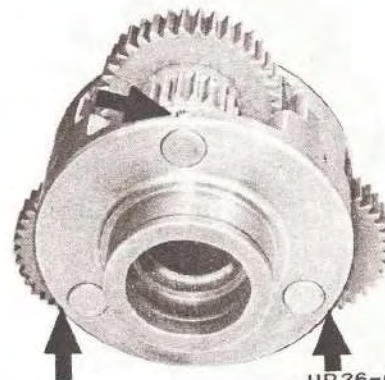
UR26-1014

- 14 Remove crawler gear annulus and pull out deep groove ball bearing.



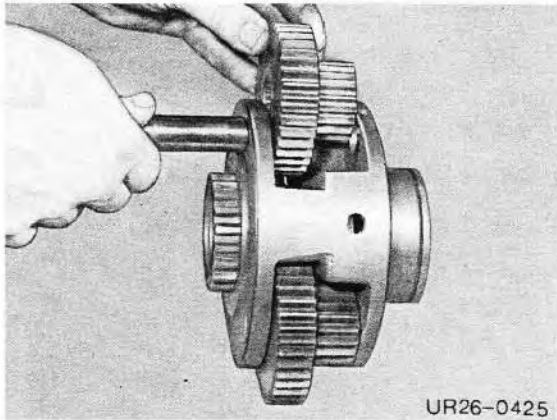
UR26-0426/1

- 15 Release bolt and screw out.



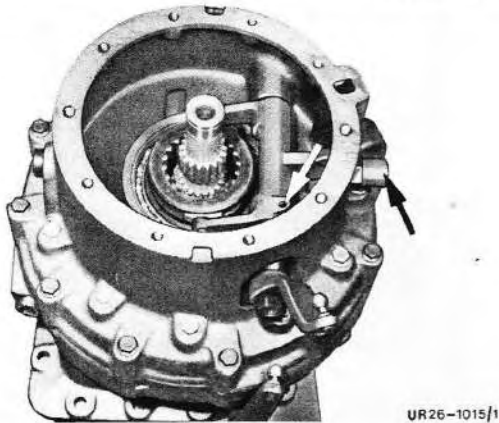
UR26-0424/1

SA 35 736

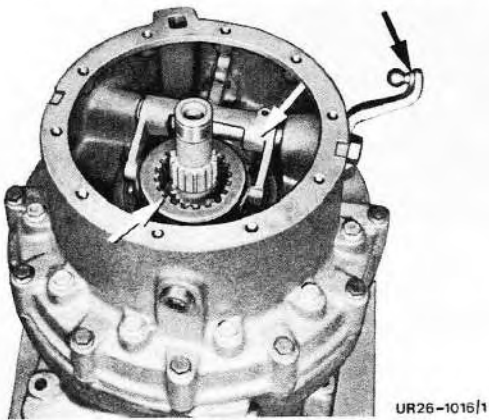


16 Remove planet shaft and gear.

Note: Watch out for friction washer, needle bearing and distance piece.



17 Remove dowel pin and interlock.

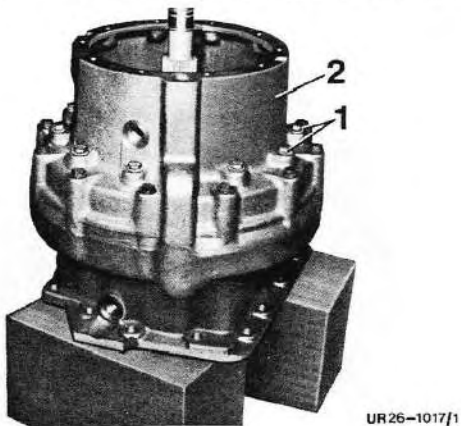


18 Remove selector shaft, selector fork with pads and sliding sleeve.

Note: Input shaft will fall downwards.

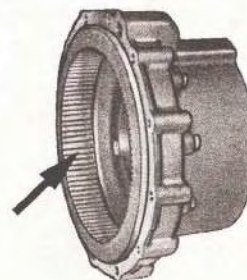
19 Remove input shaft.

20 Dismantle input shaft, see 14.1/1.



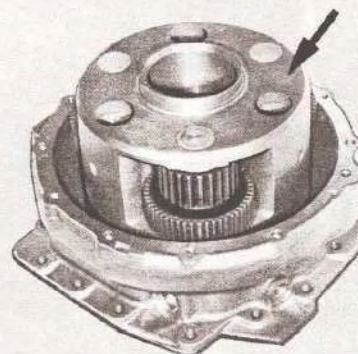
21 Loosen fastening bolts for annulus (1) and remove crawler gear casing (2).

- 22 Remove crawler gear annulus.



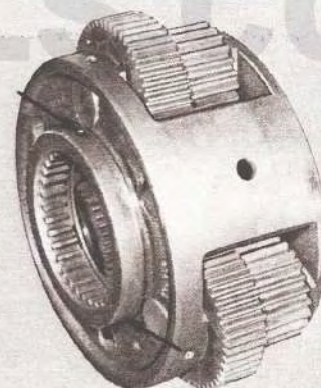
UR26-1018/1

- 23 Remove low ratio planetary gearing.



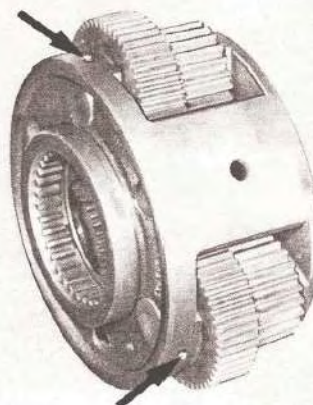
UR26-1028/1

- 24 Unscrew bolts and remove ring gear.
Applicable from transmission No. 003 168.



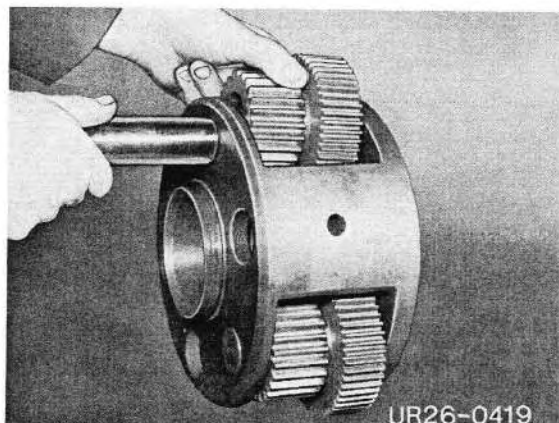
UR26-0428/1

- 25 Release bolt and unscrew.



UR26-0428/2

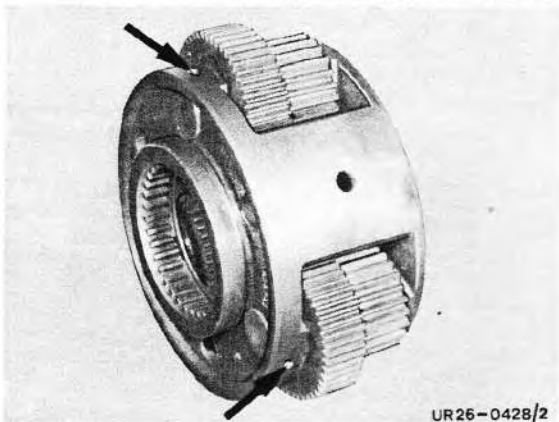
SA 35 736



26 Remove planet shaft and gear.

Note: Watch out for friction washers, needle bearing and distance pieces.

27 Clean all parts, check and replace if necessary. Blow out oil drillings and pipes with compressed air.

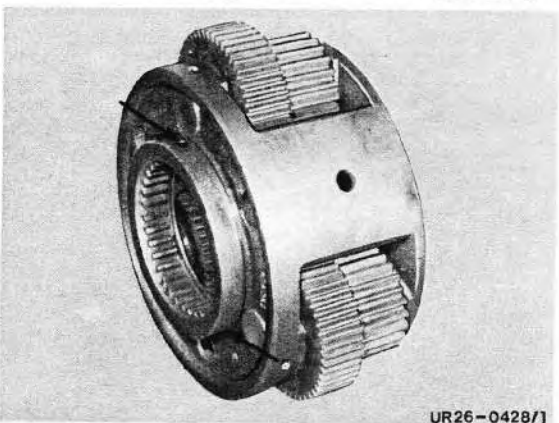


Assembly

1 Fit planet gear and shaft together with friction washers, needle rollers and distance piece.

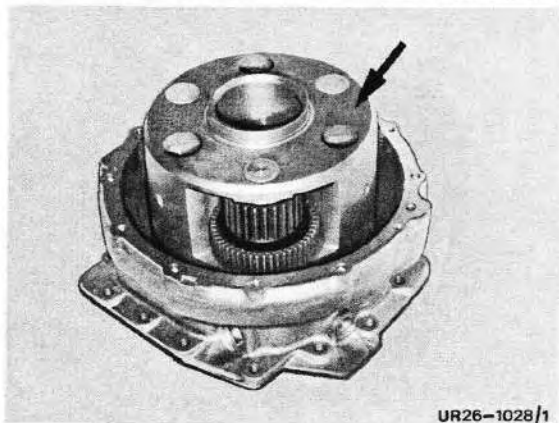
Note: Coat needle rollers with grease No. 5. Needle rollers can only be changed in sets (50 in each).

2 Screw in bolt and secure by tapping with a chisel.



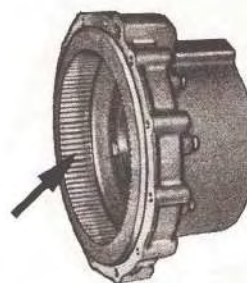
3 Fit ring gear, secure with bolts and press on large deep groove ball bearing.

Note: Coat bolts with locking material No. 1.



4 Fit low ratio planetary gearing.

- 5 Fit annulus for low ratio gears.

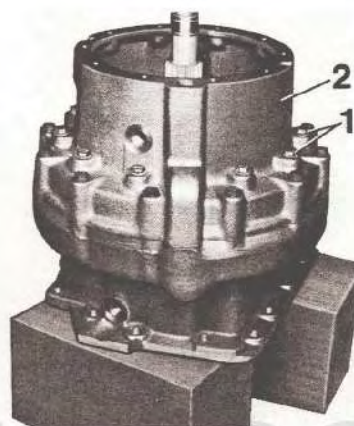


UR26-1018/1

- 6 Fit crawler gear housing with gasket and tighten bolts for crawler gear housing (2) and annulus (1). Permissible tightening torque, see 1.7/1.

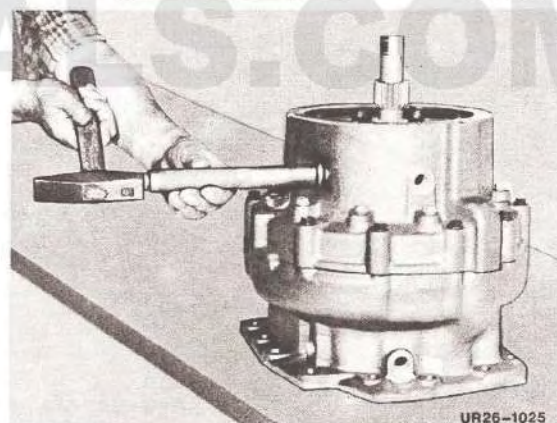
- 7 Assemble input shaft, see 6.1/1.

- 8 Fit input shaft.



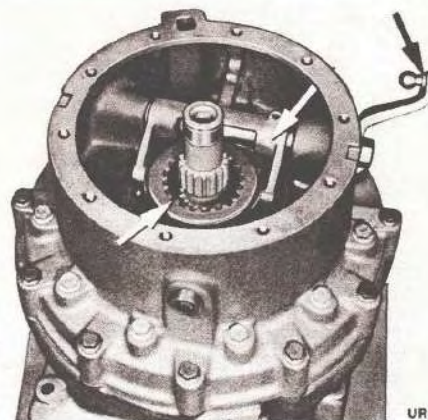
UR26-1017/1

- 9 Coat shaft seal with grease No. 3 and fit with special tool No. 16 a.



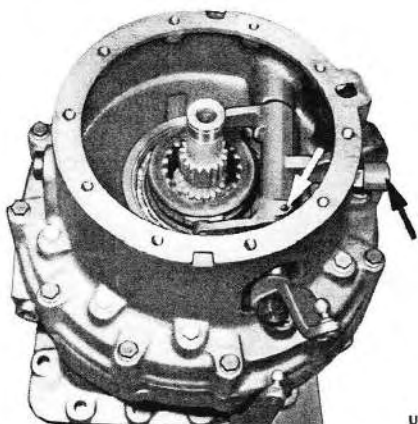
UR26-1025

- 10 Fit selector fork with pads, sliding sleeve and selector shaft according to the markings.



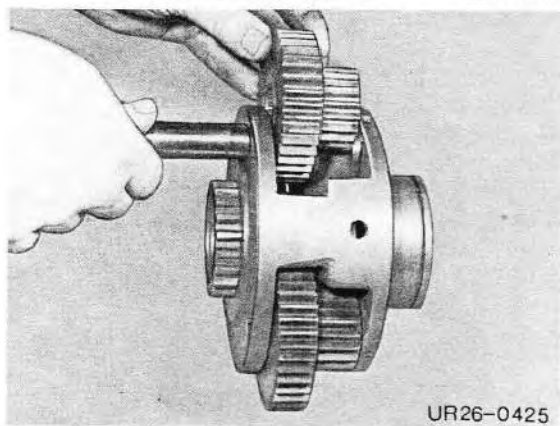
UR26-1016/1

SA 35 736



UR26-1015/1

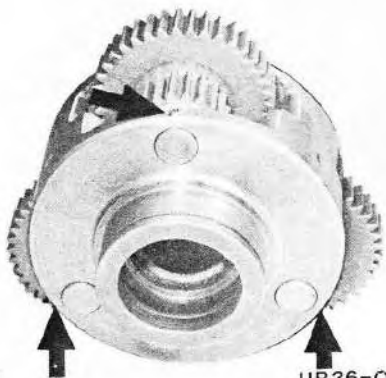
11 Fit dowel pin and interlock.



UR26-0425

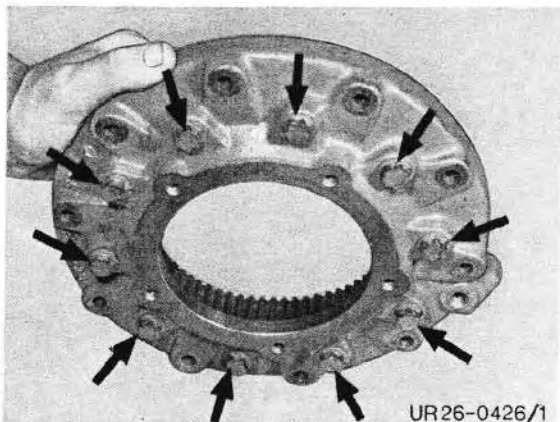
12 Fit planet gear and shaft with friction washers, needle bearings and distance piece.

Note: Coat needle bearings with grease No. 5.



UR26-0424/1

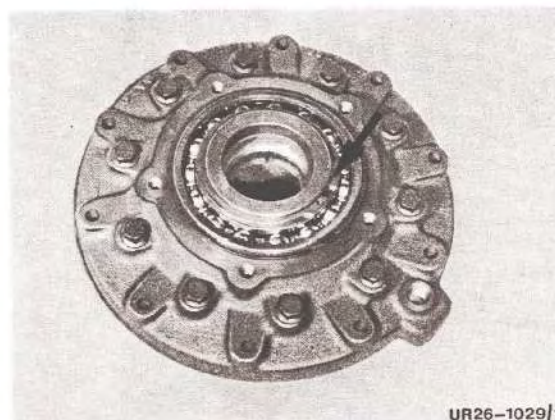
13 Screw in bolt and secure with centre punch.



UR26-0426/1

14 Fit crawler gear annulus and large deep groove ball bearing with circlip.

15 Press bearing cover onto planetary train - crawler group and secure.

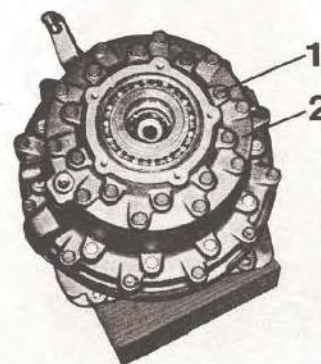


UR26-1029/1

16 Install planetary train for crawler group together with gasket and tighten bolts for crawler group (2) and ring gear (1).

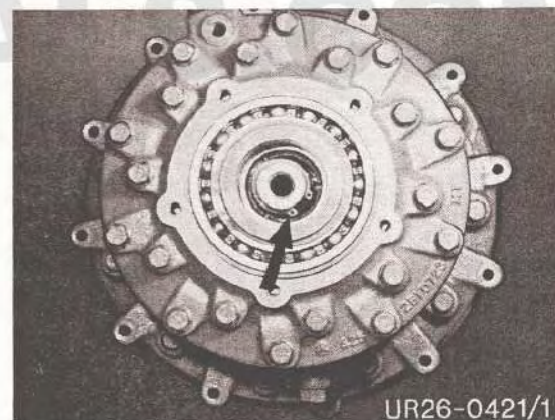
For permissible tightening torques, see 1.7/1.

Note: From chassis end No. 102 873 normal bolts are installed as standard in place of locking bolts.



UR26-1023/1

17 Install circlip.



UR26-0421/1

18 Press in small deep-groove ball bearing.

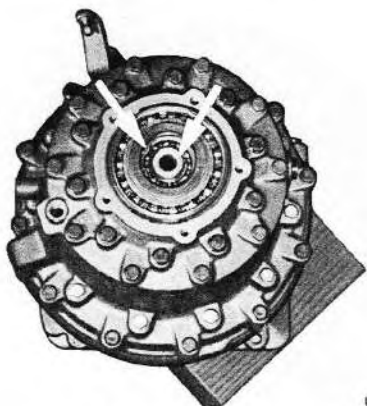
19 Press back input shaft.



UR26-1013/1

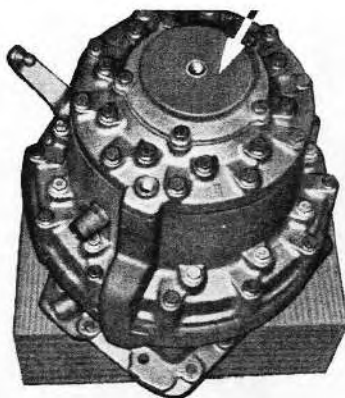
26.13 Disassembly and Assembly of Rear-Mounted Auxiliary Transmission

SA 35 736



UR26-1011/1

20 Install inner and outer retaining ring.

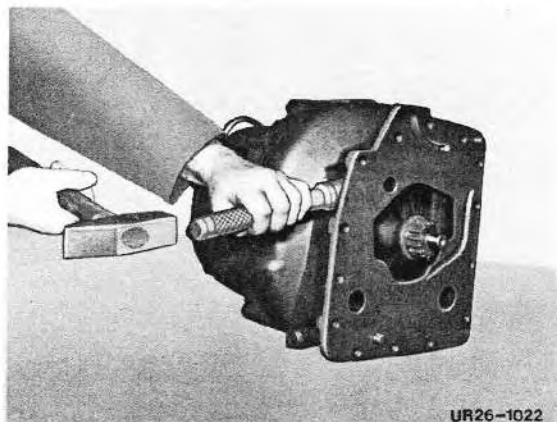


UR26-1010/1

21 Fit end cover with gasket.

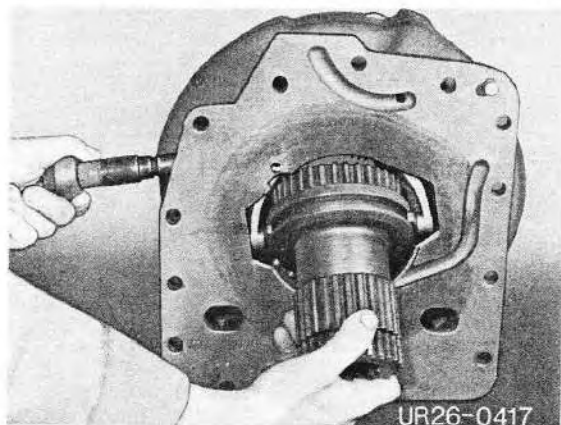
22 Attach oil pipe at end cover crawler group.

23 Turn rear-mounted auxiliary transmission through 90°.



UR26-1022

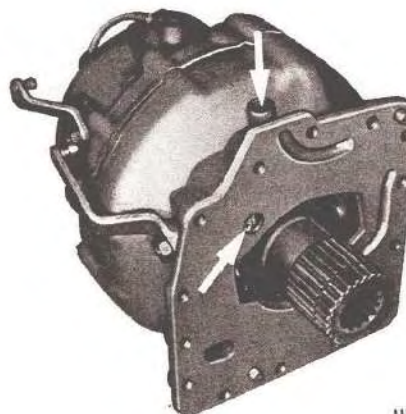
24 Coat shaft sealing ring with lubricating grease No. 3 and install using special tool No. 16 a.



UR26-0417

25 Install selector fork with fulcrum pads, sliding collar and shifting shaft according to marking.

- 26 Fit dowel pin and interlock.

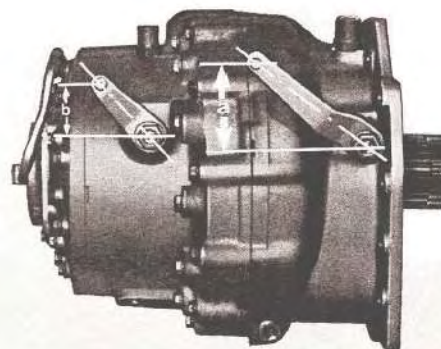


UR26-1009/1

- 27 Select lower ratio and crawler gears once and check the individual lever positions. If necessary, release nut and adjust selector lever, tighten and secure.

Measurements **a** and **b**, see 1.5/1.

- 28 Install auxiliary transmission with gasket.



UR26-1008/1

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Contents

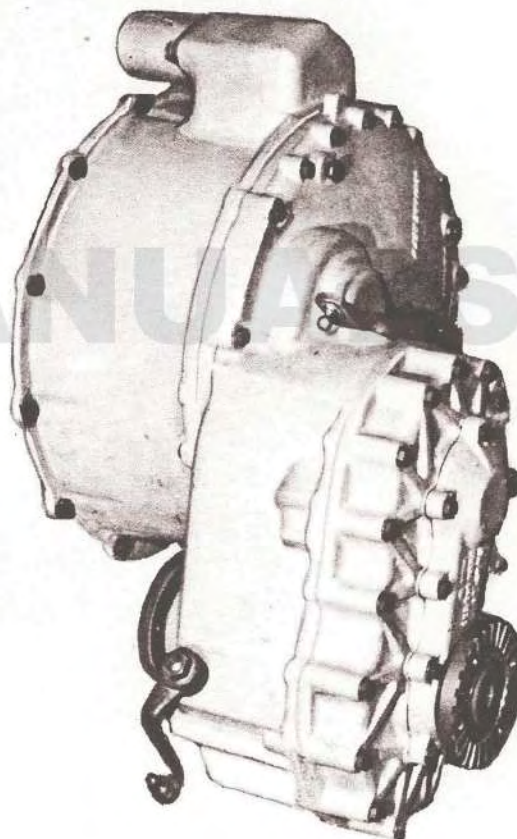
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Sectional view of engine-mounted pto shaft	1.2/1
Capacities	1.3/1
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2 Removal and installation of pto transmission	2.1/1
3 Disassembly and assembly of pto transmission	3.1/1
4 Disassembly and assembly of input shaft	4.1/1
5 Disassembly and assembly of output shaft	5.1/1

Installation Survey

Chassis		PTO transmission 540/1000/ min			
Model	Sales designation	Version	PTO output flange	Direction of rotation	Installation
435.110	U 1700	Engine and transmission PTO shaft	to front and rear	clockwise	SA 35 737
.111	U 1700 L				
.113 ¹⁾	U 1700 L/ 38				
.115	U 1300 L				
.117	U 1300 L/ 37				

1) Only transmission driven PTO shaft with PTO output flange to the front end available

General View



UR26-0611

PTO transmission 540/1000/ min

26.20 General

SA 35 737

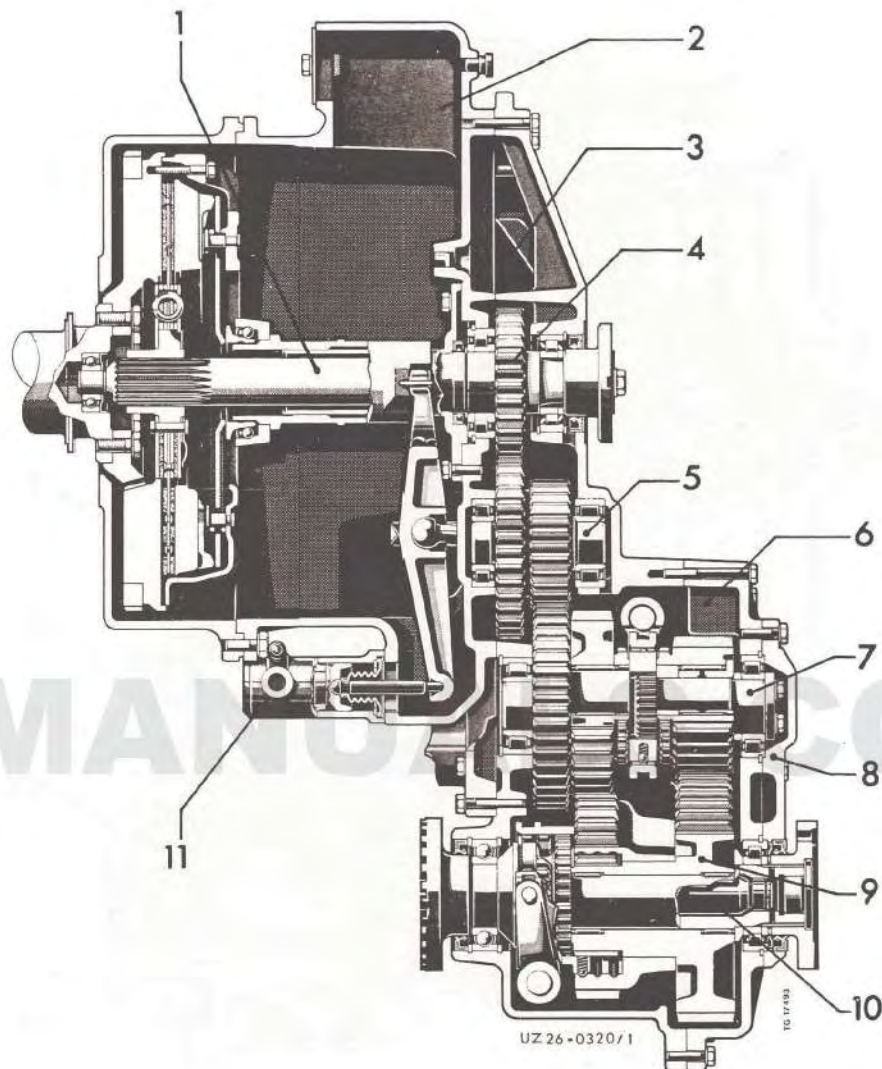
Engine Speeds, Transmission Ratios

Chassis		PTO shafts				Transmission ratios i	
Model	Sales designation	Engine speed 1/min		PTO shaft speed 1/min		540/min	1000/min
435.110	U 1700	2330	(2460)	540	(1000)	4,32	2,46
.111	U 1700 L			650	1140		
.113 ²⁾	U 1700 L/ 38						
.115	U 1300 L	2200 2800	(2320)	540	(1000)	4,08 ¹⁾	2,32 ¹⁾
.117	U 1300 L/ 37			690	1200		

1) From chassis end No. 072 513

2) Only transmission PTO shaft with PTO output at front available

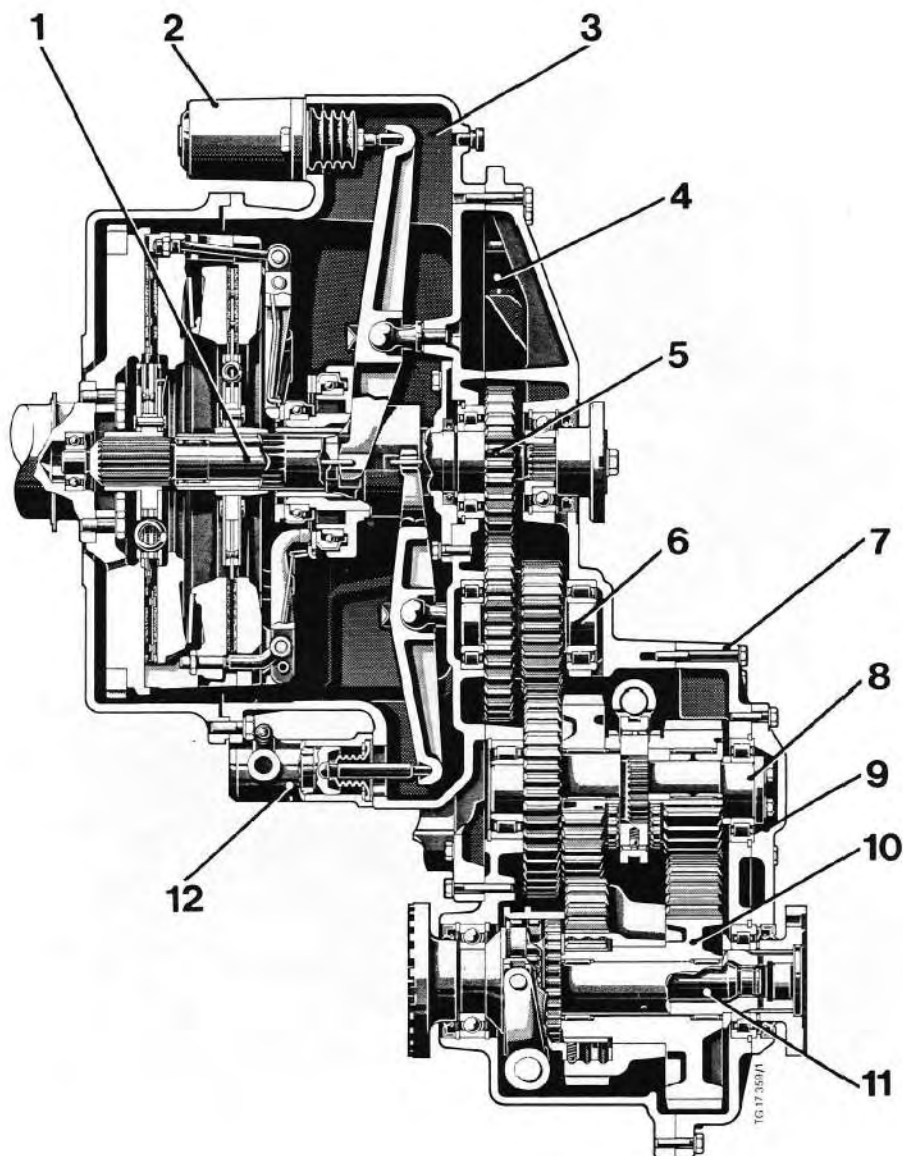
Sectional view



PTO transmission shaft, power take-off front and rear 540/1000/min

- | | | | |
|---|----------------------|----|-------------------|
| 1 | Input shaft | 7 | Input shaft |
| 2 | Clutch housing | 8 | Housing end cover |
| 3 | Transmission housing | 9 | Gear |
| 4 | Output gear | 10 | Input shaft |
| 5 | Intermediate shaft | 11 | Slave cylinder |
| 6 | Housing cover | | |

Sectional view



Engine- and PTO transmission shaft, power take-off front and rear 540/1000/min

- | | | | |
|---|----------------------|----|-------------------|
| 1 | Input shaft | 7 | Housing cover |
| 2 | Slave cylinder | 8 | Input shaft |
| 3 | Clutch housing | 9 | Housing end cover |
| 4 | Transmission housing | 10 | Gear |
| 5 | Output gear | 11 | Input shaft |
| 6 | Intermediate shaft | 12 | Slave cylinder |

Capacities

Description	Service product	SAE rating	Capacity (L)
Pto transmission	Transmission oil	80	5,75

Special tools

Cons. No.	Description	Special tool	Tool set
1	Basic drift for special tools	406 589 04 15 00	B
2	Pressing tool for shaft seal at transmission output flange with tool No. 1	435 589 01 43 00	B
3	Pressing tool for shaft seal on transmission input with tool No. 1	425 589 09 43 00	B
4	Slingshot eye bolt for removing pto transmission	435 589 01 62 00	B
5	Basic drift for special tools	385 589 00 15 00	C
6	Pressing tool for needle bearings and spacer ring in the guide flange	425 589 03 15 00	C
7	Fitting tool for shaft seal in the selector shaft drilling with tool No. 4	425 589 07 43 00	C
8	Fitting tool for shaft seal in the guide flange with tool No. 1	425 589 10 43 00	C

Consumibles

Cons. No.	Description	Part No.
1	Loctite No. 573 ¹⁾	001 989 45 20
2	Long-life grease	regular
3	Molybdenum disulphide running-in paste	regular
4	Loctite No. 241 ²⁾	002 989 70 71
5	High temperature grease	regular

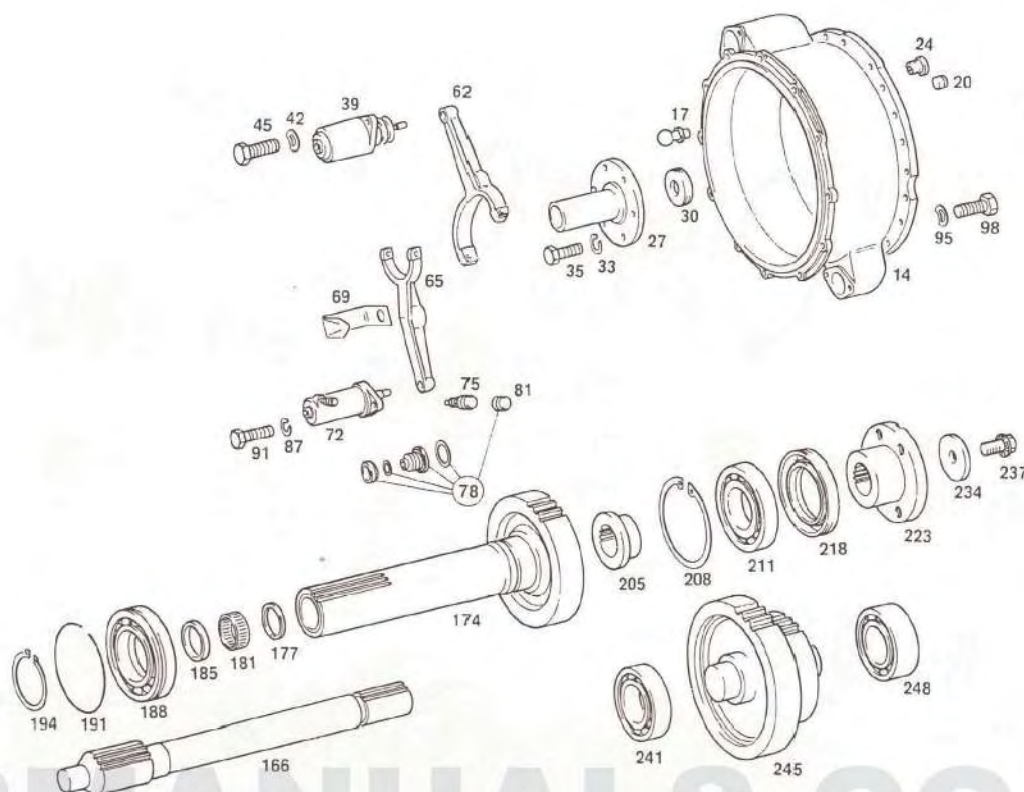
1) Surface sealant

2) Bolt locking material

Tightening torques

Description	Thread	Nm
Fastening pto output flange	M 30 x 1.5 —	300
Body support for rear cab mounting	M 20 x 1.5 — 8.8	410
Rear engine mountings on left and right of frame	M 14 x 1.5 — 10.9	140 ± 20
Fastening main transmission output flange	M 10 x 30 — 12.9	92
Pto transmission to flywheel housing	M 10 x 30 — 8.8	42
Retaining plate for input shaft	M 8 x 30 — 12.9	45
Transmission housing to clutch housing	M 8 x 45 — 8.8	21
Housing cover to transmission housing	M 8 x 65 — 8.8	21
	M 8 x 30 — 8.8	21
End cover to housing cover	M 8 x 30 — 8.8	21
Guide flange to clutch housing	M 8 x 25 — 8.8	21
Slave cylinder to clutch housing	M 8 x 25 — 8.8	21

Exploded view

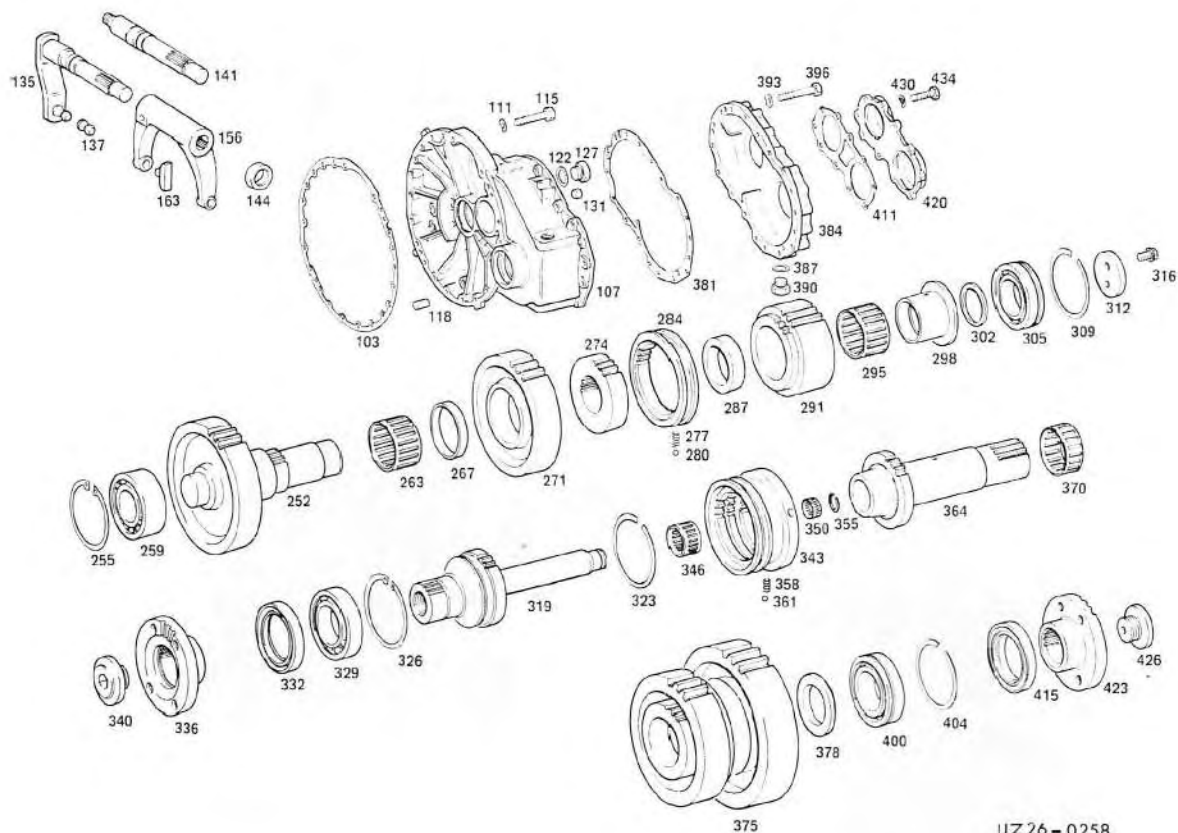


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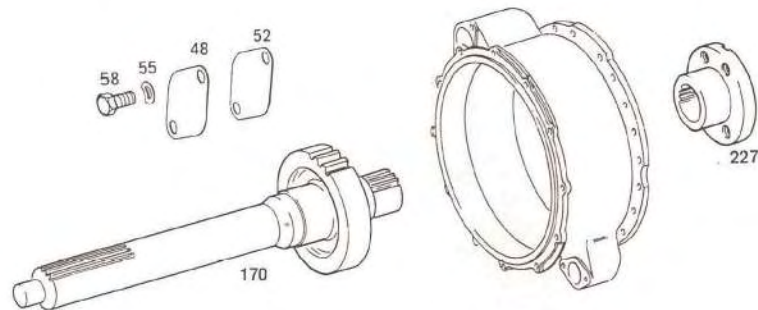
Engine- and PTO transmission shaft

- | | | | |
|----|---|-----|----------------------------|
| 14 | Clutch housing | 166 | Input shaft |
| 17 | Ball pin | 174 | Input gear |
| 20 | Plug | 177 | Spacer ring |
| 24 | Breather | 181 | Needle bearing |
| 27 | Pilot | 185 | Oil seal |
| 30 | Oil seal | 188 | Needle bearing |
| 33 | Spring ring | 191 | Snap ring |
| 35 | Screw | 194 | Locking ring |
| 39 | Slave cylinder | 205 | Follower ring |
| 42 | Spring washer | 208 | Locking ring |
| 45 | Screw | 211 | Radial bearing |
| 62 | Throwout lever, only for live pto | 218 | Oil seal |
| 65 | Throwout lever for live and transmission driven pto | 223 | Flange |
| 69 | Stirup | 234 | Disc |
| 72 | Slave cylinder | 237 | Screw |
| 75 | Valve | 241 | Cylindrical roller bearing |
| 78 | Repair kit for slave cylinder I | 245 | Gear |
| 81 | Cap | 248 | Cylindrical roller bearing |
| 87 | Spring ring | | |
| 91 | Screw | | |
| 95 | Spring washer | | |
| 98 | Screw | | |

Exploded view



Exploded view



UZ26-0257

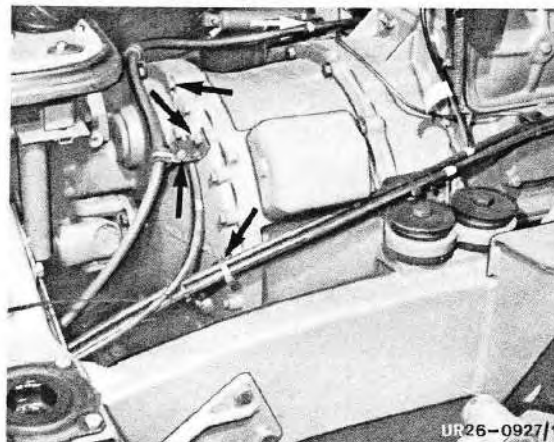
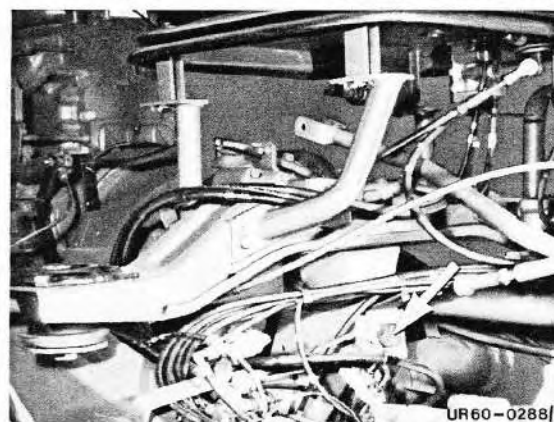
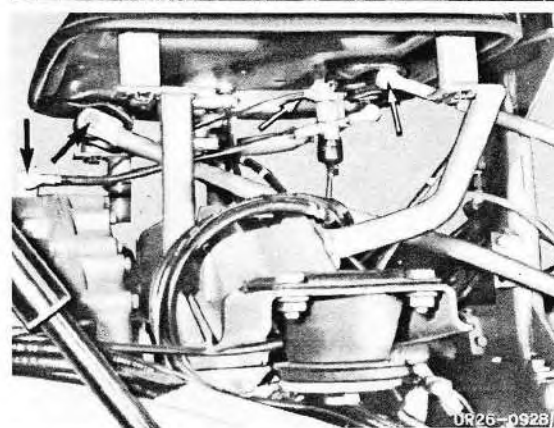
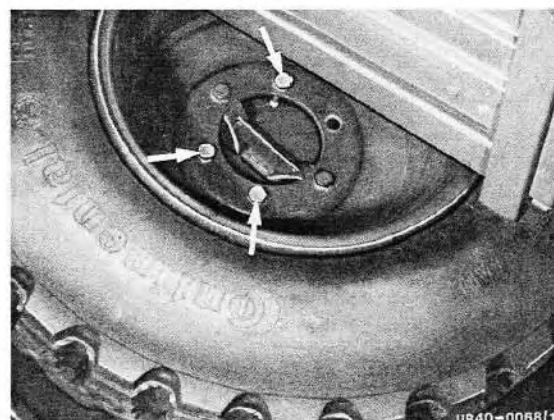
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PTO transmission shaft

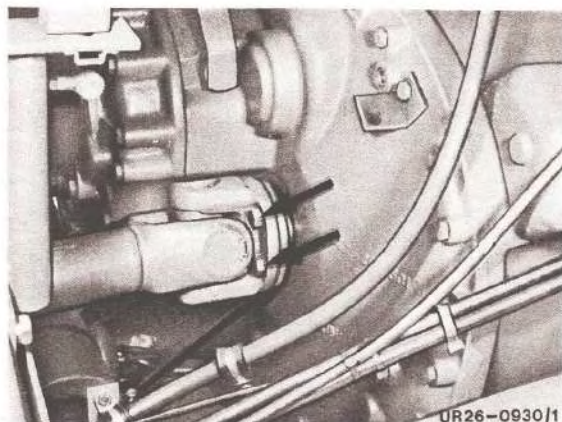
- 48 Cover plate
- 52 Gasket
- 55 Spring washer
- 58 Screw
- 170 Input shaft
- 227 Flange

Removal

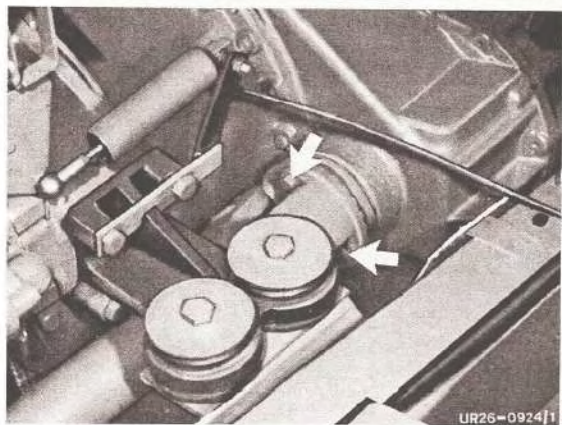
- 1 Tilt cab, see 60,5-2,1/1 or 60,5-2,2/1.
- 2 Remove spare wheel.
- 3 Unhinge main gear linkage forward/reverse gear and pto selector linkage, remove start pilot line.
- 4 Remove body mounting support and push backwards.
- 5 Disconnect main battery cable.



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- 6 Remove transmission propellor shaft.



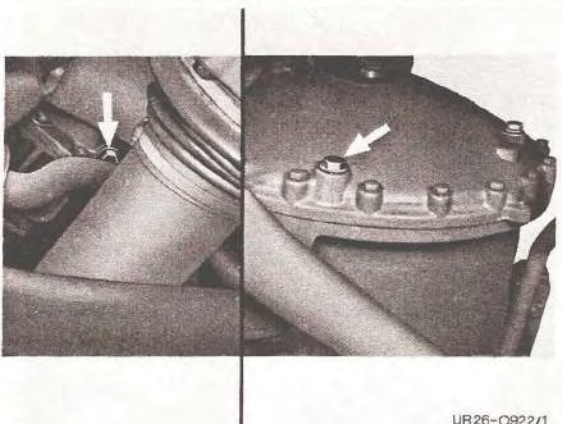
- 7 Remove front pto shaft.

Note: With front and rear pto shafts, remove both.



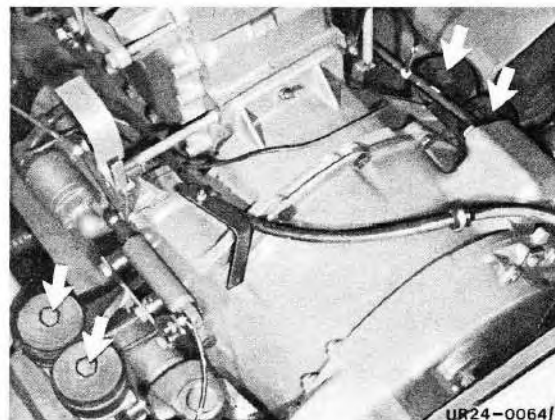
- 8 Remove clutch slave cylinder.

Note: Do not unscrew hydraulic lines.

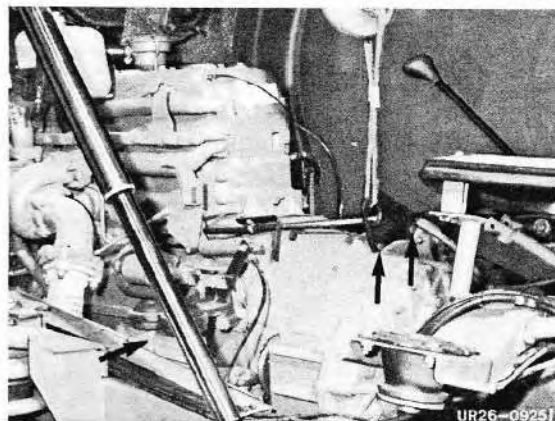


- 9 Drain oil.

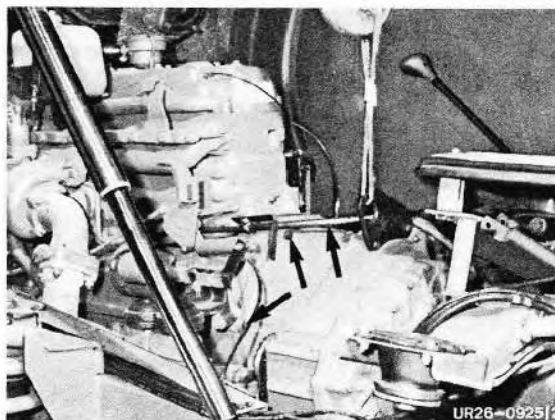
- 10 Unscrew left and right engine mounting bolts.



- 11 Fit special tool No. 4, attach pto transmission to crane, raise some way and fit wooden blocks under the left and right engine mountings.



- 12 Unbolt pto transmission, move backwards and remove by lifting.

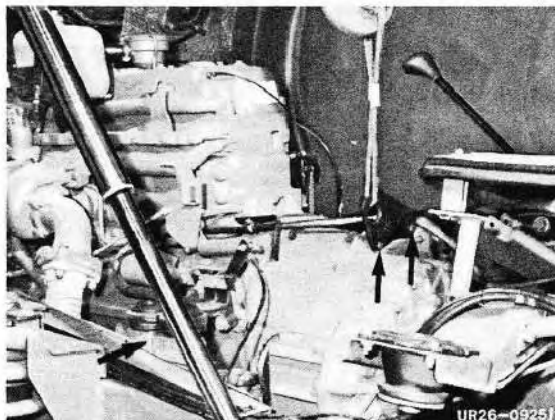


Installation

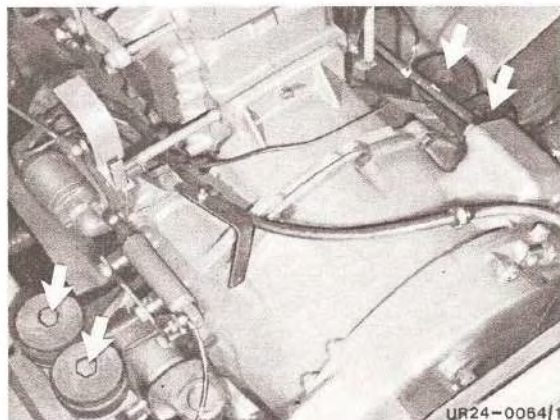
- 1 Coat pto transmission with sealant No. 1, drop down and bolt to the flywheel housing. Tightening torque, see 1.4/1.

Note: Coat bolts with locking material No. 4 and screw in with spring washers.

- 2 Lift pto transmission, remove left and right wooden blocks and unscrew special tool No. 4.

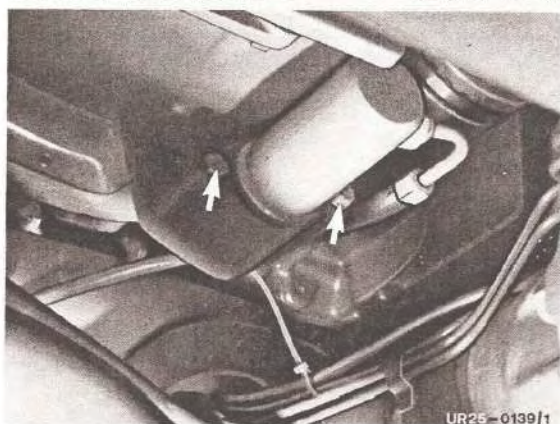


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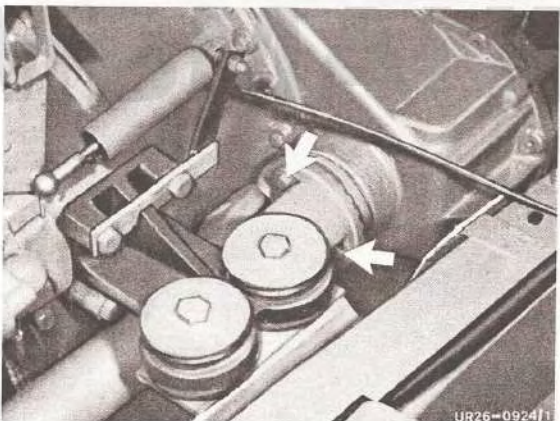
- 3 Screw in left and right engine mounting bolts and tighten.

Tightening torque, see 1.4/1.



- 4 Coat clutch slave cylinder with sealant No. 1 and bolts with locking material No. 4 and fit.

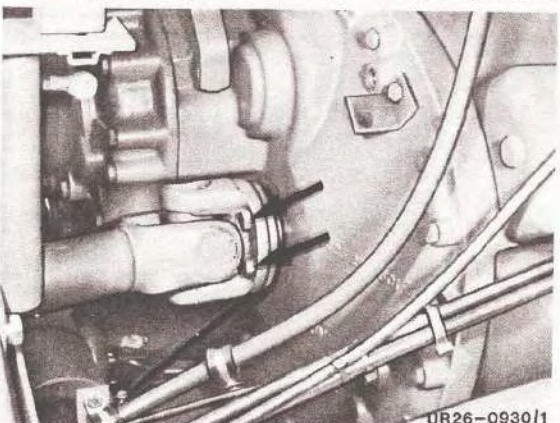
Tightening torque, see 1.4/1.



- 5 Fit front pto shaft and tighten.

Tightening torque, see 1.4/1.

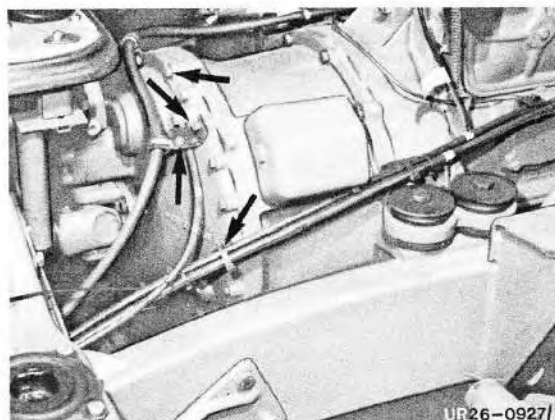
Note: With front and rear pto shafts, fit both.



- 6 Fit transmission propellor shaft and tighten.

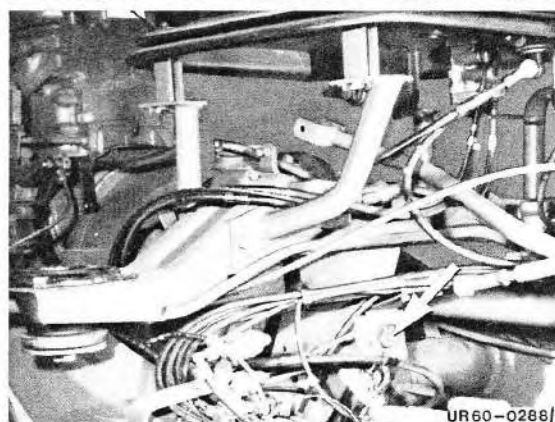
Tightening torque, see 1.4/1.

- 7 Attach main battery cable.

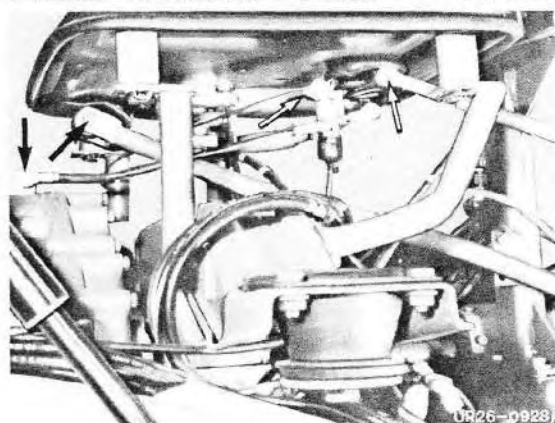


- 8 Fit body mounting and tighten.
Tightening torque, see 1.4/1.

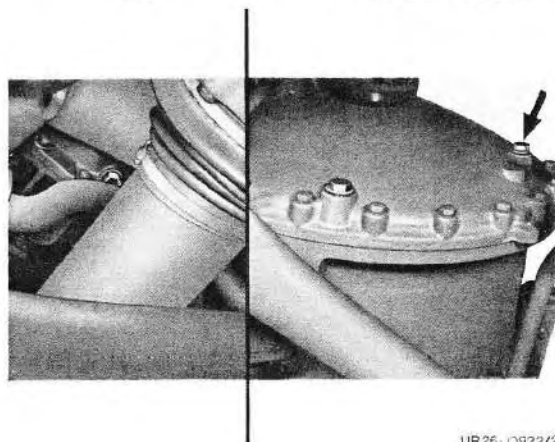
Note: Self-locking nuts must only be used once.



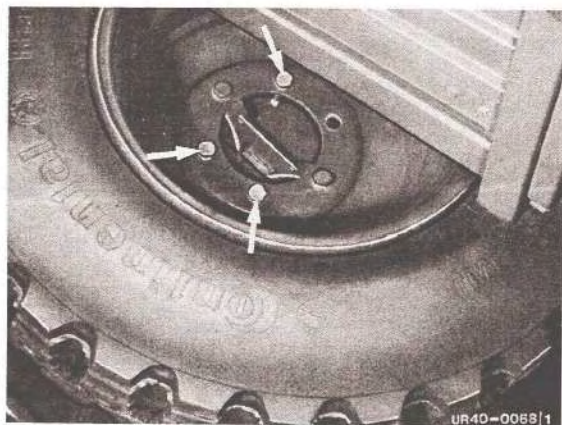
- 9 Hinge main gear linkage, forward/reverse gear
and pto selector linkage and fit start pilot line.



- 10 Fill with oil.
Capacity, see 1.3/1.



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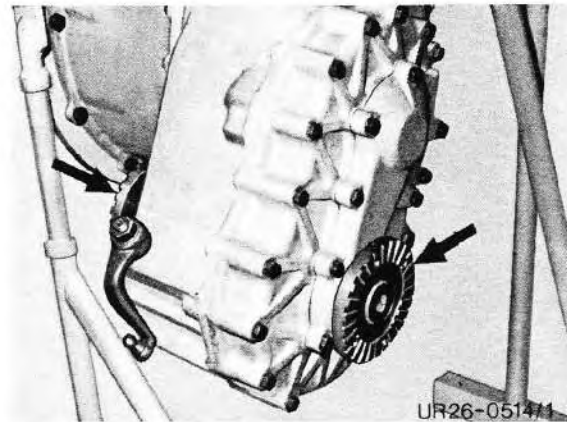


- 11 Fit spare wheel.
- 12 Lower cab, see 60,5-2,1/1 or 60,5-2,2/1.

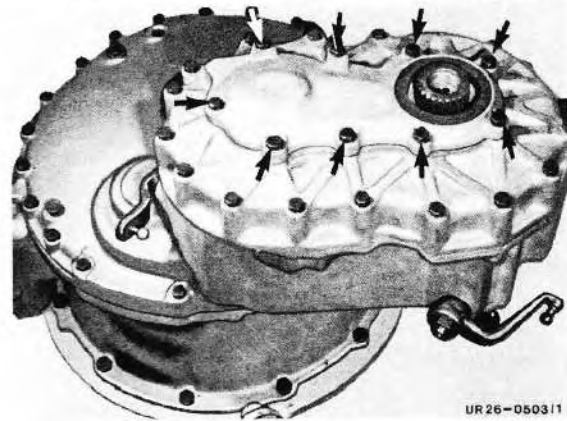
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Disassembly

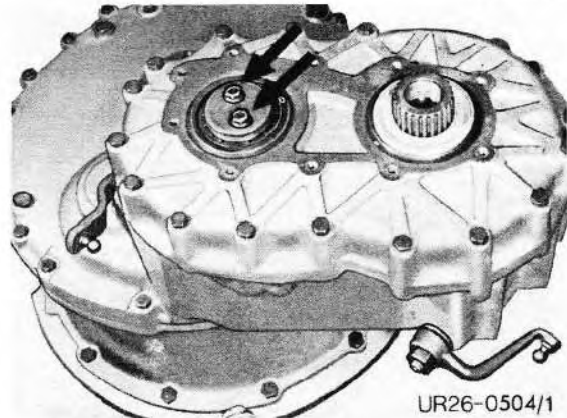
- 1 Remove pto transmission, see 2.1/1.
- 2 Clamp pto transmission in assembly stand.
- 3 Remove flange for front and rear pto shafts.
- 4 Turn pto transmission through 90°.



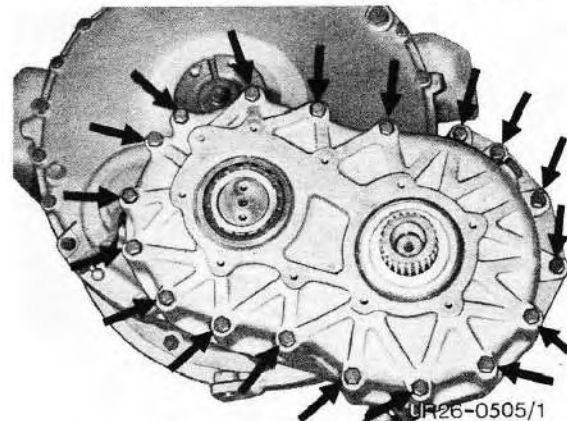
- 5 Remove end cover and, if necessary, remove shaft seal.



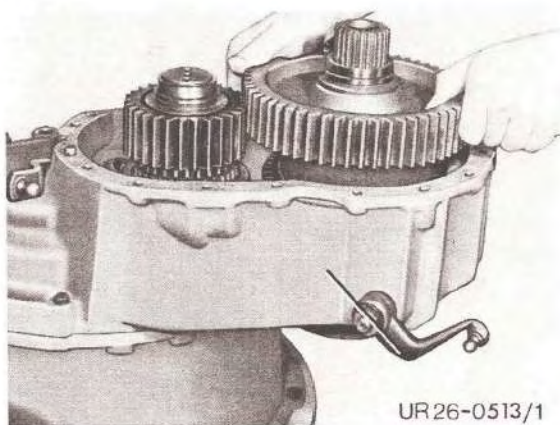
- 6 Unscrew retaining plate.



- 7 Take off housing cover and remove roller bearing.

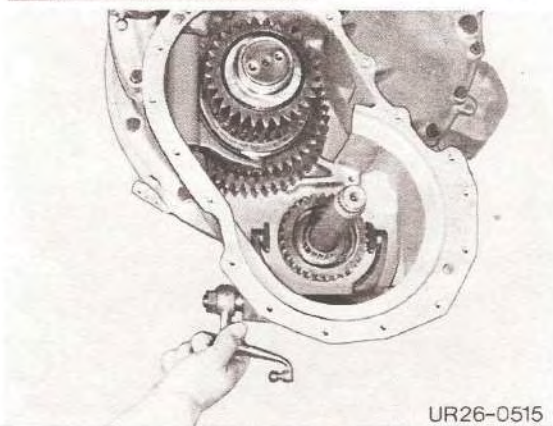


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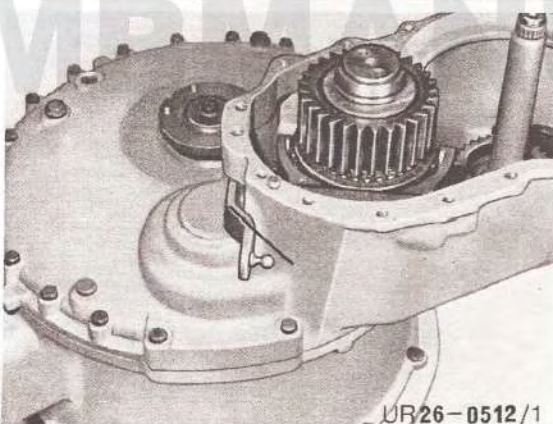
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- 8 Set the sliding sleeve in position for front pto shaft, mark and remove output gears. Disassemble output shaft, see 4.1/1.



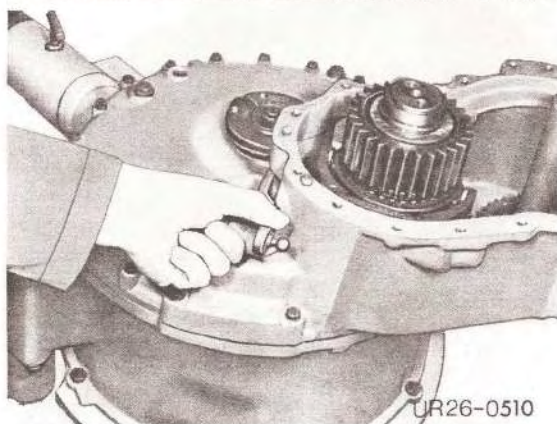
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- 9 Remove dowel pin, remove selector shaft.



UR26-0512/1

- 10 Put selector shaft on input shaft in neutral, mark and remove dowel pin.



UR26-0510

- 11 Remove selector shaft.